Extension I — Euler Substitution for Radicals

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 01 / 05

Turn a radical into a rational variable

For $s=\sqrt{x^2+a^2}$, introduce

$$t=x+s>0.$$

Then $s=t-x$ and $s^2=x^2+a^2$, so

$$x=\frac{t^2-a^2}{2t},\qquad s=\frac{t^2+a^2}{2t},\qquad \frac{dx}{s}=\frac{dt}{t}.$$

This is an Euler substitution. It proves the standard form rather than merely quoting it:

$$\int\frac{dx}{\sqrt{x^2+a^2}}=\ln t+C=\ln\left(x+\sqrt{x^2+a^2}\right)+C.$$

The absolute value is unnecessary here because $x+\sqrt{x^2+a^2}>0$ for every real x.

Euler's substitution removes the radical and preserves dx/s

Quantity	Rational form in t	Consequence
t	x+sqrt(x^2+a^2)	t is positive
x	(t^2-a^2)/(2t)	x is rational in t
s	(t^2+a^2)/(2t)	radical is rationalised
dx/s	dt/t	integral becomes logarithmic

READ THE FIGURE

The identity dx/s=dt/t is the key simplification. It follows by differentiating x(t) and comparing with s(t), not by treating dx as a separate number.

The same idea works on a branch of a difference of squares

For $x>a>0$, put $t=x+\sqrt{x^2-a^2}>0$. Now

$$x=\frac{t^2+a^2}{2t},\qquad \sqrt{x^2-a^2}=\frac{t^2-a^2}{2t},$$

and again $dx/s=dt/t$. Thus

$$\int\frac{dx}{\sqrt{x^2-a^2}}=\ln t+C.$$

On $x<-a$, the compact answer uses $\ln|x+\sqrt{x^2-a^2}|$ and is checked separately on that interval.

WORKED EXAMPLE

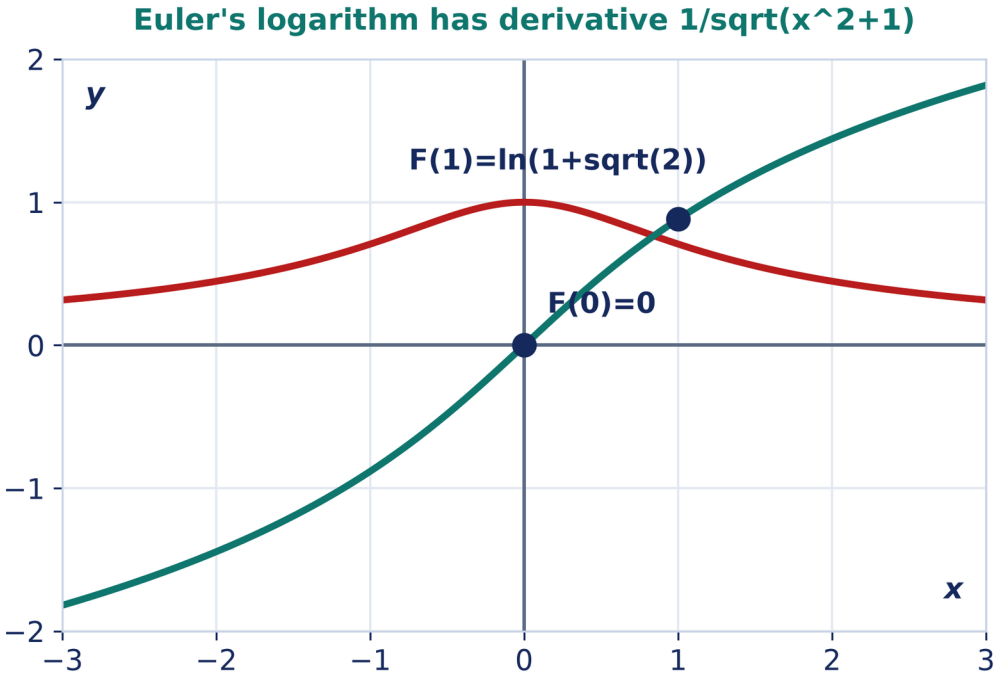
An initial value fixes the logarithmic constant

If $F'(x) = 1/\sqrt{x^2+4}$ and $F(0) = 0$, then

$$F(x) = \ln\left(x + \sqrt{x^2+4}\right) + C.$$

Since the value at zero is $\ln 2 + C$, we need $C = -\ln 2$:

$$F(x) = \ln\left(\frac{x+\sqrt{x^2+4}}{2}\right).$$



READ THE FIGURE

Red is f=1/sqrt(x^2+1), and teal is F=ln(x+sqrt(x^2+1)). F is odd, passes through zero and has positive slope f everywhere.

LESSON 02 / 05

Lower the difficulty one exponent at a time

A reduction formula expresses an integral with exponent n in terms of a smaller exponent. It is a recurrence of antiderivatives, with a constant fixed only at the end.

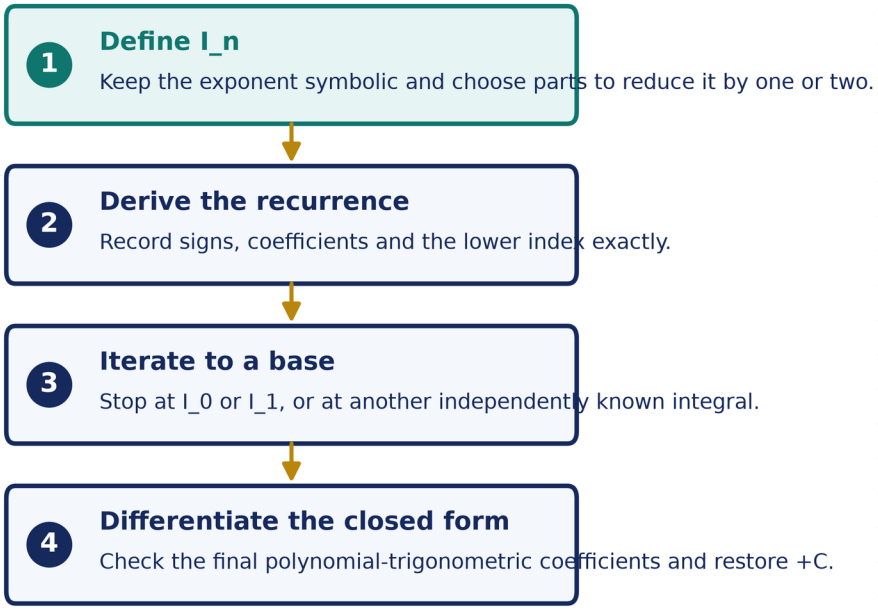
For $I_n = \int x^n e^x dx$, integration by parts gives

$$I_n = x^n e^x - nI_{n-1}, \qquad I_0 = e^x + C.$$

Thus $I_1 = e^x(x-1)$ and $I_2 = e^x(x^2 - 2x + 2)$, followed by

$$I_3 = e^x(x^3 - 3x^2 + 6x - 6) + C.$$

A reduction recurrence terminates at a known base case



READ THE FIGURE
A recurrence is not just a computational shortcut: it also proves the antiderivative's structural form by induction on the exponent.

Logarithmic powers follow the same pattern

For $x > 0$, let $J_n = \int (\ln x)^n dx$. With $u = (\ln x)^n$ and $dv = dx$,

$$J_n = x(\ln x)^n - nJ_{n-1}, \qquad J_0 = x + C.$$

The first three closed forms are

$$J_1 = x\ln x - x, \qquad J_2 = x(\ln x)^2 - 2x\ln x + 2x,$$

$$J_3 = x(\ln x)^3 - 3x(\ln x)^2 + 6x\ln x - 6x,$$

all up to the same final additive constant.

Trigonometric powers step down by two

For $n \geq 2$, let $S_n = \int \sin^n x\,dx$. By parts,

$$S_n = -\tfrac{1}{n}\sin^{n-1} x\cos x + \tfrac{n-1}{n}S_{n-2},$$

with $S_0 = x + C$ and $S_1 = -\cos x + C$. In particular,

$$S_6 = -\tfrac{1}{6}\sin^5 x\cos x - \tfrac{5}{24}\sin^3 x\cos x - \tfrac{5}{16}\sin x\cos x + \tfrac{5}{16}x + C.$$

Differentiating this expression returns $\sin^6 x$. The source's Wallis questions S15, S19, Q30 and Q43 apply limits to this same recurrence in Chapter 21.

COMMON TRAP

The recurrence's constant belongs to the completed answer

The displayed recurrence is an equality of chosen antiderivatives. When iterating, keep a consistent choice and add the arbitrary constant at the end. Do not allow unrelated constants at each stage to masquerade as extra terms.

Extension III — Parameter Differentiation

NOTEBOOK FLOW: read straight down the page, then continue on the next sheet.

LESSON 03 / 05

A parameter can manufacture a logarithm

For $x>0$ and $a\neq -1$, start with

$$H(a,x)=\int x^a dx=\frac{x^{a+1}}{a+1}+C.$$

Holding x fixed and differentiating with respect to a brings down $\ln x$:

$$\frac{\partial H}{\partial a}=\int x^a \ln x\,dx=\frac{x^{a+1}((a+1)\ln x-1)}{(a+1)^2}+C.$$

At $a=0$ this reproduces $\int \ln x\,dx=x\ln x-x$. At $a=1$ it gives

$$\int x\ln x\,dx=\frac{x^2}{2}\ln x-\frac{x^2}{4}+C,$$

which is the antiderivative used in the source's S14 and Q28.

Parameter differentiation creates a logarithmic hierarchy

Starting object	Operation in a	New integrand
x^(a+1)/(a+1)	first derivative	x^a ln x
x^(a+1)/(a+1)	second derivative	x^a (ln x)^2
a=0 after first	specialise	ln x
a=1 after first	specialise	x ln x

READ THE FIGURE

Differentiate first with a general, then substitute the parameter value. Setting a=0 before differentiating would erase the parameter being used.

A second differentiation gives squared logarithms

Differentiating once more,

$$\int x^a(\ln x)^2dx=x^{a+1}\left(\frac{(\ln x)^2}{a+1}-\frac{2\ln x}{(a+1)^2}+\frac{2}{(a+1)^3}\right)+C.$$

At $a=0$,

$$\int (\ln x)^2dx=x(\ln x)^2-2x\ln x+2x+C,$$

in agreement with the reduction formula J_2 in Extension II. This is the indefinite form behind P14 and Q29; their endpoint limits belong to Chapter 21.

WORKED EXAMPLE

Cross-check two methods at a=2

Parameter differentiation gives

$$\int x^2 \ln x \, dx = x^3\left(\frac{\ln x}{3} - \frac{1}{9}\right) + C.$$

Direct by parts with $u = \ln x$ and $dv = x^2 dx$ gives exactly the same expression. Their agreement checks both the parameter derivative and the by-parts coefficient.

COMMON TRAP

State the domain and the order of operations

Here $x > 0$ makes $x^a = e^{a \ln x}$ unambiguous. Differentiate with respect to a while x is fixed, and only then specialise a. The method generates an antiderivative; any later definite limit requires its own convergence check.

LESSON 04 / 05

Some antiderivatives require new named functions

An elementary function is built from algebraic, exponential, logarithmic, trigonometric and inverse-trigonometric operations. Liouville's theory shows that several innocent-looking integrands have no antiderivative within that class.

This does **not** say that no antiderivative exists. It says that we must enlarge the function library.

Special functions can be introduced by derivative and normalisation

Integrand	Named antiderivative	Defining derivative
exp(-x^2)	(sqrt(pi)/2) erf(x)	erf'(x)=2 exp(-x^2)/sqrt(pi)
sin x / x	Si(x)	Si'(x)=sin x/x
1 / ln x	li(x)	li'(x)=1/ln x
exp(x) / x	Ei(x)	Ei'(x)=exp(x)/x

READ THE FIGURE
Together with erf(0)=0 and Si(0)=0, the first two derivatives determine smooth special functions. The logarithmic and exponential integrals require domain branches around their singularities.

For example,

$$\int e^{-x^2}dx = \frac{\sqrt{\pi}}{2}\operatorname{erf}(x) + C, \qquad \operatorname{erf}'(x) = \frac{2}{\sqrt{\pi}}e^{-x^2}.$$

Similarly, after assigning the continuous value 1 to $\sin x/x$ at $x = 0$,

$$\int \frac{\sin x}{x}dx = \operatorname{Si}(x) + C, \qquad \operatorname{Si}(0) = 0.$$

These statements are verified by the same principle as every other antiderivative: differentiate the named function.

The frontier is about representation, not computational failure

The source's Gaussian S16/Q32/Q46 and Dirichlet Q37 questions concern **definite** values. Chapter 21 will show that some of those values are exact even though no elementary antiderivative is available. A definite symmetry or double-integral method can succeed where an elementary formula for $F(x)$ does not exist.

COMMON TRAP

Never write 'the integral does not exist' merely because the usual methods fail

The precise statement is "there is no elementary antiderivative." Numerical approximation, a special function, a convergent series or an exact definite-integral technique may still be available. Conversely, quoting a special-function name without its derivative and domain is not a verification.

LESSON 05 / 05

One complex exponential integrates sine and cosine together

Using $i^2 = -1$ and $e^{ix} = \cos x + i\sin x$,

$$\int e^{(a+ib)x}dx = \frac{e^{(a+ib)x}}{a+ib} + C, \qquad a+ib \neq 0.$$

Taking real and imaginary parts gives the general formulas

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C,$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C.$$

At $a = b = 1$, the sine formula becomes Q18's $\frac{1}{2}e^x(\sin x - \cos x)$.

An ansatz gives the same result without complex arithmetic

Try

$$F = e^{ax}(A\sin bx + B\cos bx).$$

Then

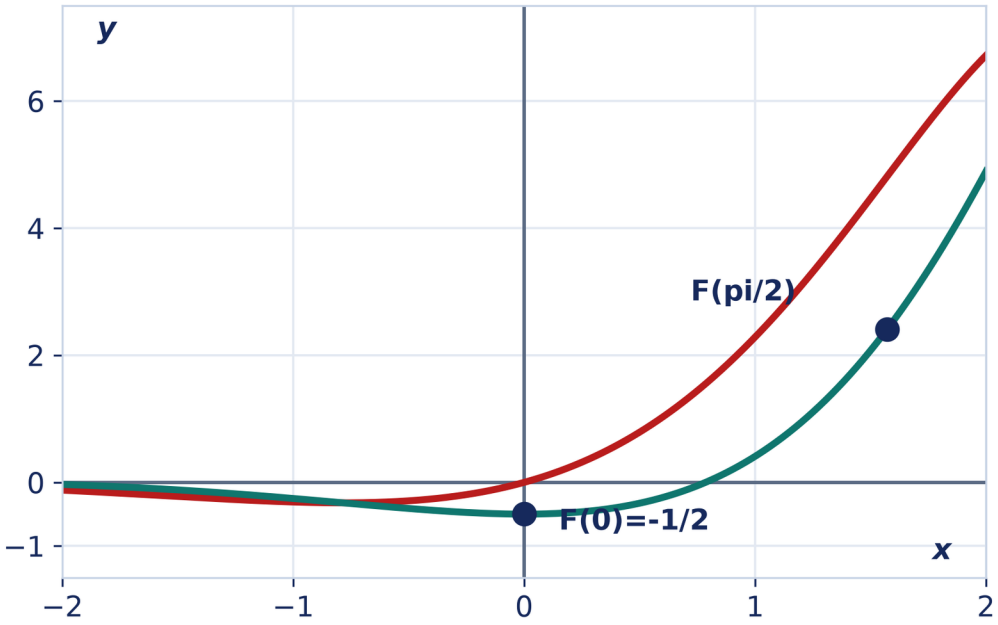
$$F' = e^{ax}((aA - bB)\sin bx + (bA + aB)\cos bx).$$

To make $F' = e^{ax}\sin bx$, solve

$$aA - bB = 1, \qquad bA + aB = 0,$$

which gives $A = a/(a^2+b^2)$ and $B = -b/(a^2+b^2)$. The cosine case only changes the right-hand sides of these two linear equations.

The cyclic antiderivative has slope e^x sin x



READ THE FIGURE

Red is f=e^x sin x; teal is F=e^x(sin x-cos x)/2. The slope of teal at zero is f(0)=0, while at pi/2 it is f(pi/2)=e^(pi/2), about 4.8105.

WORKED EXAMPLE

The same ansatz handles a damped cosine

With $a=-2$ and $b=3$,

$$\int e^{-2x}\cos 3x\,dx=\frac{e^{-2x}(-2\cos 3x+3\sin 3x)}{13}+C.$$

Differentiation restores both exponential damping and the frequency 3; the denominator is $(-2)^2+3^2=13$.

COMMON TRAP

Do not mix the sine and cosine coefficient rows

The sine antiderivative has $a\sin bx-b\cos bx$; the cosine antiderivative has $a\cos bx+b\sin bx$. A quick differentiation at one general parameter set is safer than relying on visual similarity between the two formulas.